

9-11

Theorem 5.1 If $\vec{b} \in \text{range } A$, then the general solution to $A\vec{x} = \vec{b}$ is given by the set

$$\{\vec{x}_p\} + \text{null } A = \{ \vec{x}_p + \vec{x} : A\vec{x}_p = \vec{b} \text{ and } A\vec{x} = \vec{0} \}$$

↙
particular soln

General solution: $\vec{x} = \vec{x}_p + \sum x_i \vec{v}_i$ where $\{\vec{v}_1, \dots, \vec{v}_k\}$ are a basis for null A .

Exercise:

1) Find the line of intersection of the planes

$$3x_1 - x_2 + 2x_3 = 5 \text{ and } x_1 + 3x_3 = 6$$

$$\left[\begin{array}{ccc|c} 3 & -1 & 2 & 5 \\ 1 & 0 & 3 & 6 \end{array} \right] \xrightarrow{-3r_2} \left[\begin{array}{ccc|c} 0 & -1 & -7 & -13 \\ 1 & 0 & 3 & 6 \end{array} \right]$$

$$\begin{array}{lcl} x_2 + 7x_3 = 13 & \Rightarrow & x_1 = 6 - 3x_3 \\ x_1 + 3x_3 = 6 & & x_2 = 13 - 7x_3 \\ & & x_3 = t \end{array}$$

The general solution i.e. the equation of the line of intersection is given by

$$\vec{x} = \begin{bmatrix} 6 \\ 13 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ -7 \\ 1 \end{bmatrix}$$

Matrix Transpose (Axler 7.A, Strang 2.7)

Def 5.2 Let $A \in \mathbb{R}^{m \times n}$ be such that A_{ij} denotes the ij th component of A . Then $A^T \in \mathbb{R}^{n \times m}$ is given $A^T_{ij} = A_{j,i}$
 rows of A are the columns of A^T

Def 5.3 If $A = A^T$ then A is symmetric

Theorem 5.4 Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times l}$. Then
 $(AB)^T = B^T A^T$

Proof: As per usual let M_{ij} denote the ij th component of the matrix M .

Now as

$$\begin{aligned} (AB)^T_{ij} &= AB_{j,i} = \overbrace{(A_{j,\cdot})(B_{\cdot,i})}^{\substack{\in \mathbb{R}^n \\ \in \mathbb{R}^n}} \\ &\quad \text{inner product} \\ &= \sum_{k=1}^n A_{j,k} B_{k,i} \end{aligned}$$

$$\begin{aligned}
&= \sum_{k=1}^n B_{k,i} A_{j,k} \\
&= \sum_{k=1}^n B_{i,k}^T A_{k,j}^T \\
&= (B^T)_{i,\cdot} (A^T)_{\cdot,j} \\
&= (B^T A^T)_{i,j}
\end{aligned}$$

we have that $(AB)^T = B^T A^T$

Theorem 5.5 Properties of the transpose

- i. If $A\vec{x} = \vec{y}$, then $\vec{x}^T A^T = \vec{y}^T$
- ii. $(A^T)^T = A$
- iii. $\langle A\vec{x}, \vec{y} \rangle = \langle \vec{x}, A^T \vec{y} \rangle$ (A^T is the adjoint of A)

If $T \in \mathcal{L}(V, W)$ then the adjoint $T^* \in \mathcal{L}(W, V)$ is such that $\langle T\vec{x}, \vec{y} \rangle_W = \langle \vec{x}, T^* \vec{y} \rangle_V$

- iv. If A is invertible then A^T is also invertible with inverse $(A^T)^{-1} = (A^{-1})^T$
- v. The products AA^T and $A^T A$ are square-symmetric for any A .

$$(AA^T)^T = (A^T)^T A^T = AA^T$$

Exercise:

2) Show by example that AA^T can be invertible even when A is not.

$$\text{let } A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \text{ so that } A^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\text{and } AA^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Condition for invertibility: AA^T is invertible when $\text{null } AA^T = \emptyset$ or when

$$\text{i. } \text{null } A^T = \emptyset$$

$$\text{ii. } \text{null } A \cap \text{range } A^T = \emptyset \quad (\text{always true})$$

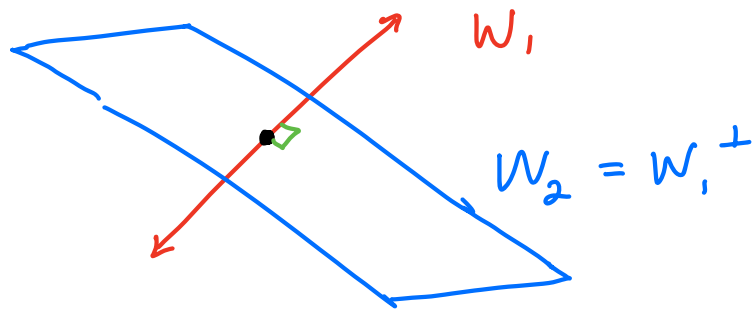
Orthogonal Compliments

Def 5.6 Given subspace W of an inner product space V , we define the orthogonal compliment of W as $(W\text{-perp})$

$$W^\perp = \{ \vec{x} \in V : \langle \vec{x}, \vec{y} \rangle = 0 \text{ for all } \vec{y} \in W \}$$

Example: In $\mathbb{R}^3$ the orthogonal compliment of line through the origin is a plane through the

origin that is orthogonal to the line.



Theorem 5.7 Let W be a subspace V .

Then $W^\perp$ is also a subspace of V with dimension $\dim W^\perp = \dim V - \dim W$. In fact $V = W \oplus W^\perp$ as $W \cap W^\perp = \emptyset$.

If $W \subseteq V$ (not nec. a subspace) then $W^\perp$ is subspace and $(W^\perp)^\perp = \text{span } W$

If W is a subspace, then $(W^\perp)^\perp = W$

Fundamental Subspaces (Strang 3.5, 4.1)

Def 5.8 Given $A \in \mathbb{R}^{m \times n}$, the Fundamental subspaces of A are: $\text{range } A$, $\text{null } A$, $\text{range } A^T$, and $\text{null } A^T$ where the latter two are subspaces of $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively.

i. $\text{range } A = \{ \vec{y} \in \mathbb{R}^m : \vec{y} = A\vec{x} \text{ for some } \vec{x} \in \mathbb{R}^n \}$
"column space of A "

$$= \text{Span} (\text{lin. ind. columns of } A)$$

$$\text{ii. range } A^T = \{ \vec{x} \in \mathbb{R}^n : \vec{x} = A^T \vec{y} \text{ for some } \vec{y} \in \mathbb{R}^m \}$$

"row space of A "

$$= \text{Span} (\text{lin. ind. rows of } A)$$

$$= \text{Span} (\text{non zero rows of } U \text{ or } R)$$

$$\text{iii. null } A = \{ \vec{x} \in \mathbb{R}^n : A \vec{x} = \vec{0} \}$$

$$\text{iv. null } A^T = \{ \vec{y} \in \mathbb{R}^m : \underbrace{A^T \vec{y}} = \vec{0} \}$$

"left null space"

$$\vec{y}^T A = \vec{0}^T$$

Theorem 5.9 $(\text{range } A)^\perp = \text{null } A^T$ and

$(\text{null } A)^\perp = \text{range } A^T$. Thus $\mathbb{R}^m = \text{range } A \oplus \text{null } A^T$

and $\mathbb{R}^n = \text{null } A \oplus \text{range } A^T$

Proof: Exercise. Pick arbitrary $\vec{y} \in \text{range } A$ and $\vec{x} \in \text{null } A^T$ and show that $\vec{x}^T \vec{y} = 0$.

Def 5.10 We will call $\text{rank } A$ the "column rank" of A and $\text{rank } A^T = \dim(\text{range } A^T)$ the "row rank" of A . If $A \in \mathbb{R}^{m \times n}$ then we say

that A has full column rank if $\text{rank } A = m$
(row) (rank $A^T = n$)

Theorem 5.11 (row rank = column rank)

Let $A \in \mathbb{R}^{m \times n}$. Then A defines an isomorphism from $\text{range } A^T$ to $\text{range } A$. Hence $\text{rank } A = \text{rank } A^T$.

Proof: Let $T \in \mathcal{L}(\text{range } A^T, \text{range } A)$ be defined by $T(\vec{x}) = A \cdot \vec{x}$. Since $\text{range } A^T \cap \text{null } A = \emptyset$, $\ker T = \emptyset$ and thus T is injective. Now let $\vec{y} \in \text{range } A$ and let $\vec{x} \in \mathbb{R}^n$ be s.t. $A\vec{x} = \vec{y}$. As $\mathbb{R}^n = \text{range } A^T \oplus \text{null } A$ let $\vec{x} = \vec{x}_1 + \vec{x}_2$ where $\vec{x}_1 \in \text{range } A^T$ and $\vec{x}_2 \in \text{null } A$. Now as

$$\vec{y} = A(\vec{x}_1 + \vec{x}_2) = A\vec{x}_1 + A\vec{x}_2 = A\vec{x}_1 = T(\vec{x}_1)$$

we have that T is surjective and hence an isomorphism.

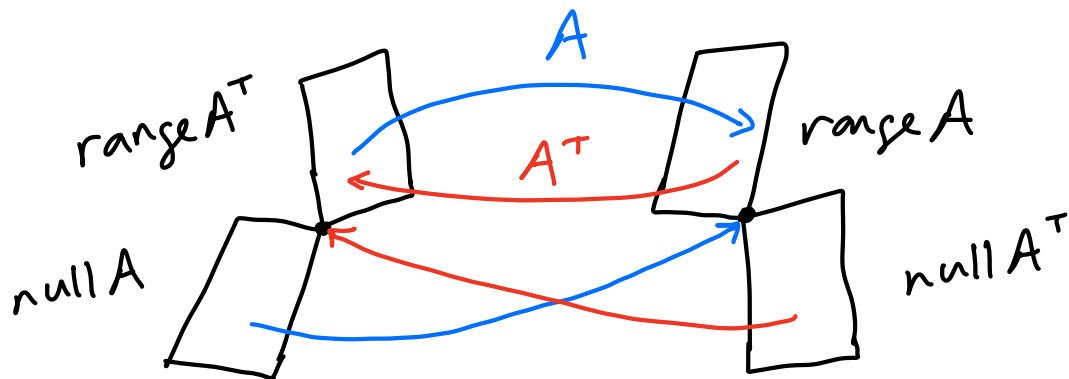
Corollary 5.12 Given $A \in \mathbb{R}^{m \times n}$ so that

$$\text{rank } A = r$$

- i. $\text{rank } A^T = r$
- ii. $\text{nullity } A = n - r$

iii. nullity $A^T = m - r$

$$A \in \mathbb{R}^{m \times n} : \mathbb{R}^n \rightarrow \mathbb{R}^m$$



Exercise

3) Find bases for the fundamental subspaces of

$$A = \begin{bmatrix} 1 & 2 & 0 & -3 \\ -3 & 1 & 2 & 9 \\ -1 & 5 & 2 & 3 \end{bmatrix}$$

$$A_{3,\cdot} = 2A_{1,\cdot} + A_{2,\cdot}$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ -3 & 1 & 2 & 9 \\ -1 & 5 & 2 & 3 \end{bmatrix} \xrightarrow[\substack{+3r_1 \\ +r_1}]{r_3 = 2r_1 + r_2 \text{ if you prefer}} \begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & 7 & 2 & 0 \\ 0 & 7 & 2 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & 1 & 2/7 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{-2r_2} \begin{bmatrix} 1 & 0 & -4/7 & -3 \\ 0 & 1 & 2/7 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \text{range } A = \text{Span} \left(\begin{bmatrix} 1 \\ -3 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} \right)$$

$$\text{range } A^T = \text{Span} \left(\begin{bmatrix} 1 \\ 2 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \\ 2 \\ 9 \end{bmatrix} \right)$$

look at the pivots!

$$= \text{Span} \left(\begin{bmatrix} 1 \\ 0 \\ -4/7 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2/7 \\ 0 \end{bmatrix} \right)$$

Setting $A\vec{x} = \vec{0}$:

$$x_1 - \frac{4}{7}x_3 - 3x_4 = 0$$

$$x_2 + \frac{2}{7}x_3 = 0$$

$$x_1 = \frac{4}{7}x_3 + 3x_4$$

$$x_2 = -\frac{2}{7}x_3$$

$$x_3 = x_3$$

$$x_4 = x_4$$

$$\text{null } A = \text{Span} \left(\begin{bmatrix} 4/7 \\ -2/7 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right)$$

$\text{null } A^T$: as $A_{3,\cdot} = 2A_{1,\cdot} + A_{2,\cdot}$

$$2A_{1,\cdot} + A_{2,\cdot} - A_{3,\cdot} = \vec{0} \quad \text{or}$$

$$2A_{\cdot,1}^T + A_{\cdot,2}^T - A_{\cdot,3}^T = \vec{0} \quad \text{and thus}$$

$$\vec{0} = A^T \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

Hence $\begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} \in \text{null } A^T$ and as

$\dim(\text{null } A^T) = 3 - 2 = 1$, we conclude that

$$\text{null } A^T = \text{Span} \left(\begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} \right)$$

One could also find $\text{null } A^T$ by eliminating

$$A^T = \begin{bmatrix} 1 & -3 & -1 \\ 2 & 1 & 5 \\ 0 & 2 & 2 \\ -3 & 9 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & 1 \\ 0 & 7 & 7 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix} \text{ ect } \dots$$

4) (Next time) Given $W = \{ \vec{x} \in \mathbb{R}^3 : 2x_1 - x_2 + 3x_3 = 0 \}$
find a matrix M s.t. $W^\perp = \text{null } M$